

CMDA-4654
Group Exercise 1 Part A
Group 17

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1. Teammate Introduction

My teammate's name is Sophia. She's from Fall's Church City, VA. Her favorite food to eat in Blacksburg is Kuma Moon and her favorite hobby is watching TV shows with her friends.

2. Data Introduction

The area of data we chose to explore is sports. The data set is sourced from the Society for American Baseball Research's Lahman Baseball Database.

The data is linked [here](#). We downloaded `FieldingPost.csv` and `batting.csv`.

summarization of what the data corresponds to

Data Dictionary

`FieldingPost.csv` contains post-season fielding data with the following variables:

Variable	Description
<code>playerID</code>	Player ID code
<code>yearID</code>	Year
<code>stint</code>	Player's stint (order of appearances within a season)
<code>teamID</code>	Team
<code>lgID</code>	League
<code>G</code>	Games
<code>AB</code>	At Bats
<code>R</code>	Runs
<code>H</code>	Hits
<code>2B</code>	Doubles
<code>3B</code>	Triples
<code>HR</code>	Home Runs
<code>RBI</code>	Runs Batted In
<code>SB</code>	Stolen Bases
<code>CS</code>	Caught Stealing
<code>BB</code>	Base on Balls
<code>SO</code>	Strikeouts
<code>IBB</code>	Intentional Walks
<code>HBP</code>	Hit by Pitch
<code>SH</code>	Sacrifice Hits
<code>SF</code>	Sacrifice Flies
<code>GIDP</code>	Grounded Into Double Plays

`batting.csv` contains batting statistics with the following variables:

Variable	Description
<code>playerID</code>	Player ID code
<code>yearID</code>	Year
<code>stint</code>	Player's stint (order of appearances within a season)
<code>teamID</code>	Team
<code>lgID</code>	League
<code>G</code>	Games
<code>AB</code>	At Bats
<code>R</code>	Runs
<code>H</code>	Hits
<code>2B</code>	Doubles
<code>3B</code>	Triples
<code>HR</code>	Home Runs
<code>RBI</code>	Runs Batted In

Variable	Description
SB	Stolen Bases
CS	Caught Stealing
BB	Base on Balls
SO	Strikeouts
IBB	Intentional Walks
HBP	Hit by Pitch
SH	Sacrifice Hits
SF	Sacrifice Flies
GIDP	Grounded Into Double Plays

3. Analysis and Discussion

Tables

Table 1: Statistics Aggregated by League (With Pitchers)

Variable	Minimum	Mean	Maximum	Standard.Deviation
At Bats	591.000	174,553.716	560,290.000	157,348.290
Runs	104.000	24,711.552	83,649.000	22,555.768
Hits	147.000	46,993.257	144,433.000	41,919.870
Home Runs	6.000	5,321.506	24,452.000	5,851.509
Stolen Bases	23.000	3,564.981	11,275.000	3,175.340
Base on Balls	53.000	17,889.640	58,519.000	16,246.039
Strikeouts	42.000	30,199.314	137,181.000	34,012.337
Batting Average	0.242	0.273	0.329	0.015

discussion

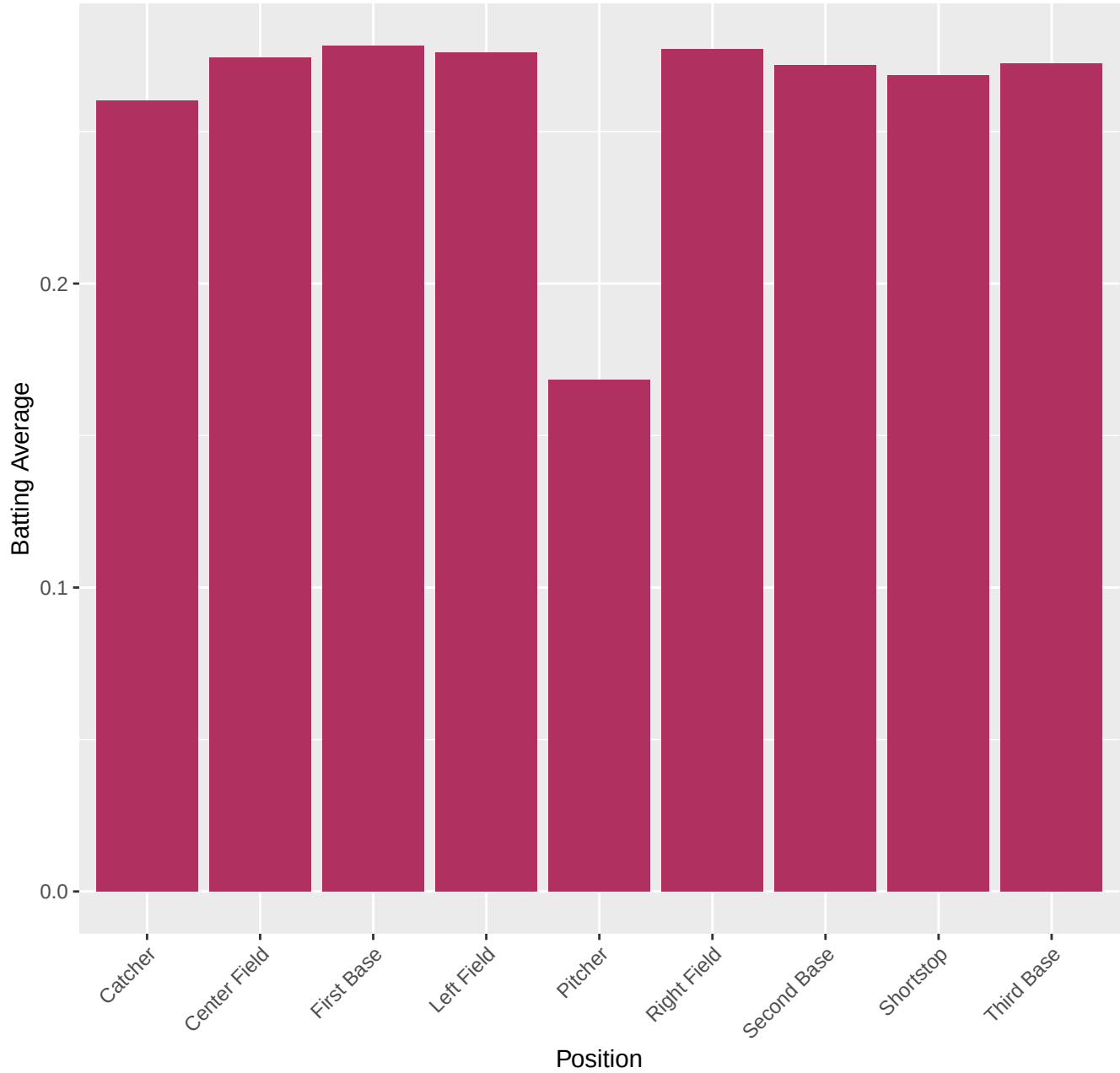
Table 2: Statistics Aggregated by League (Without Pitchers)

Variable	Minimum	Mean	Maximum	Standard.Deviation
At Bats	526.000	168,516.751	538,625.000	154,766.877
Runs	98.000	24,274.253	82,198.000	22,439.454
Hits	139.000	45,976.808	141,279.000	41,596.386
Home Runs	6.000	5,280.586	24,286.000	5,835.485
Stolen Bases	22.000	3,548.490	11,268.000	3,170.987
Base on Balls	53.000	17,593.084	57,672.000	16,154.062
Strikeouts	34.000	28,331.230	127,168.000	32,830.774
Batting Average	0.247	0.279	0.339	0.016

discussion

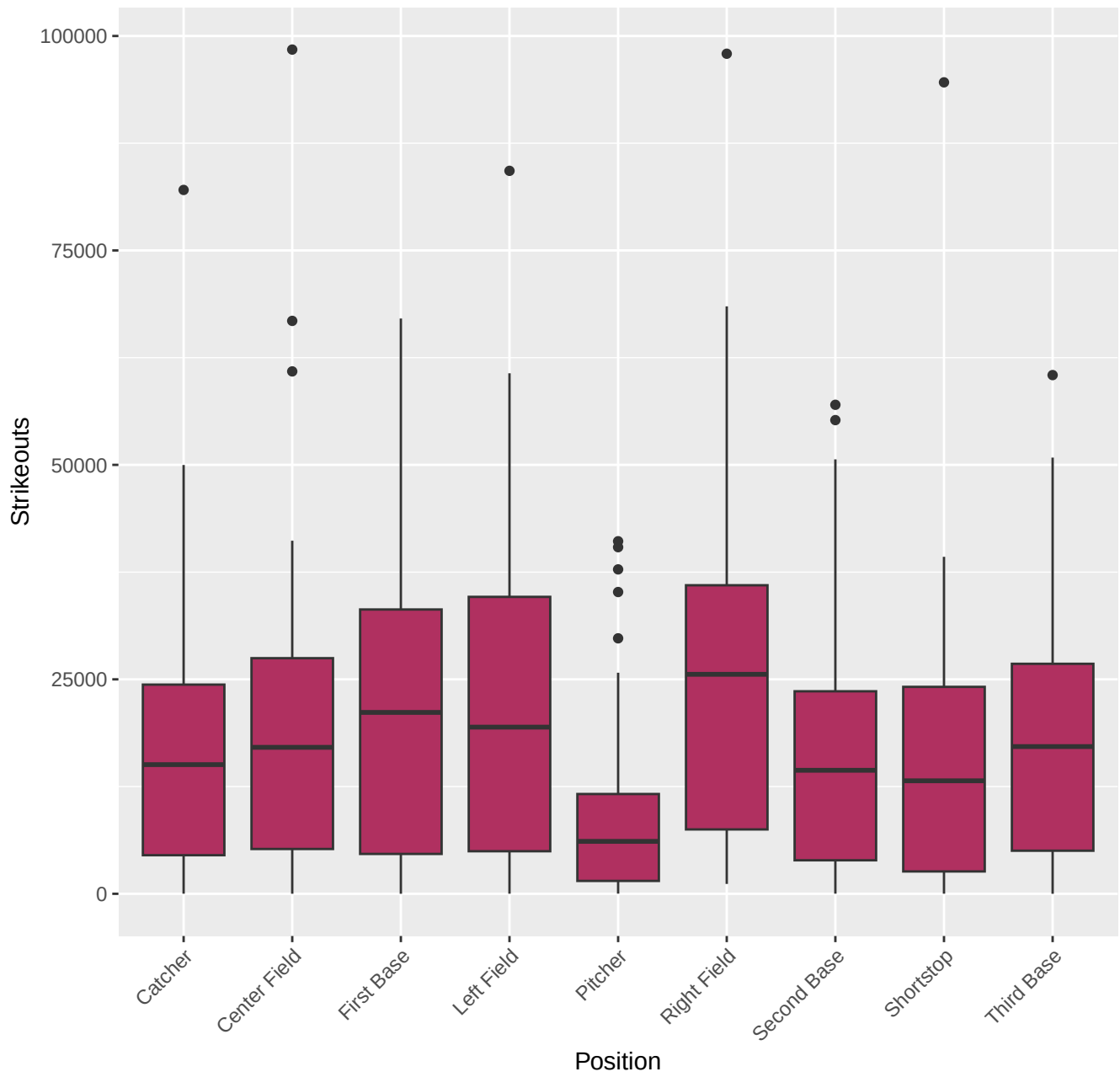
Plots

Batting Average by Position



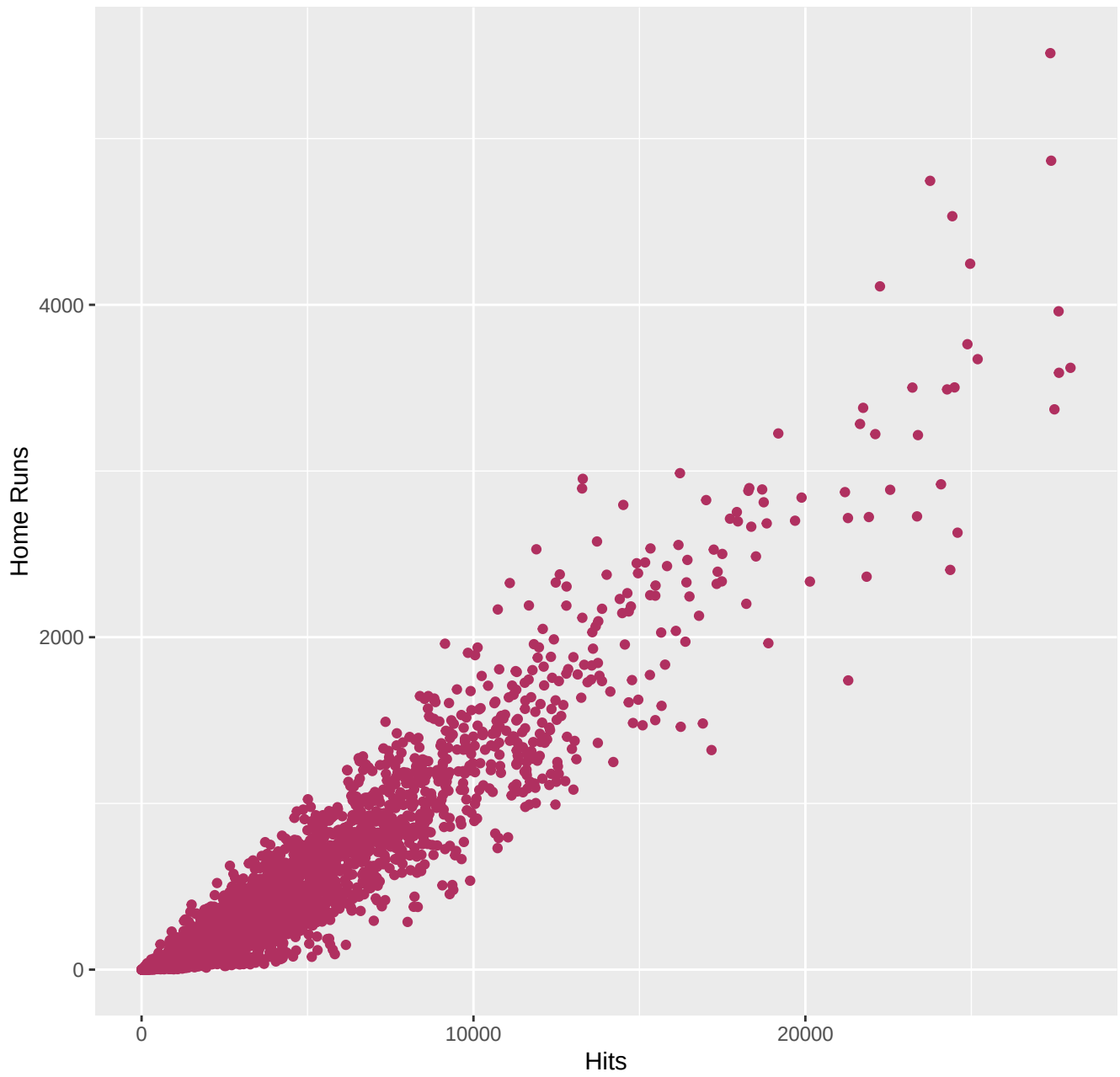
discussion

Strikeouts Across Teams by Position



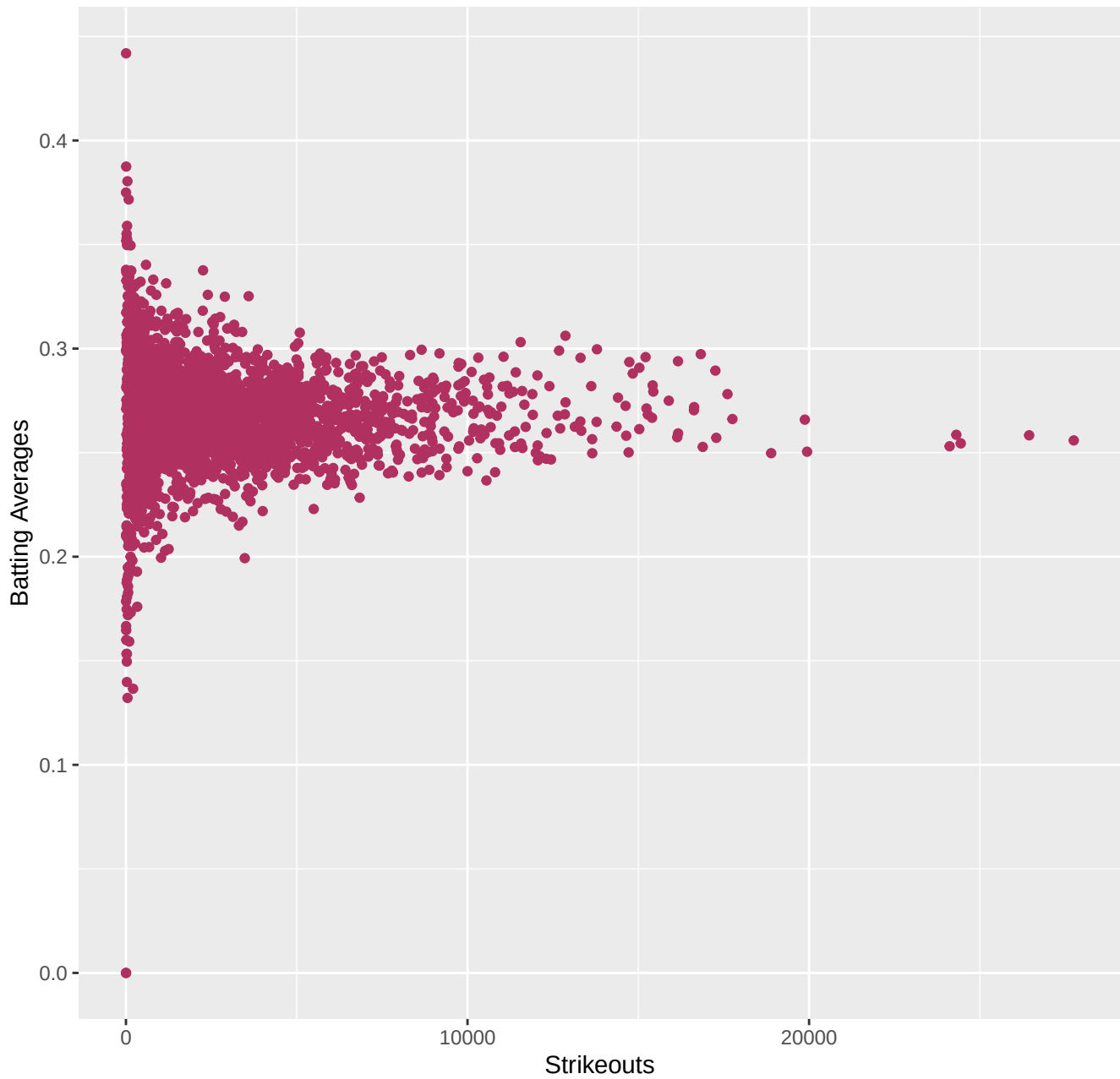
discussion

Hits vs Home Runs Across Teams



discussion

Strikeouts vs Batting Average Across Teams



4. Citations

Lahman Baseball Database – Society for American Baseball Research. (2025). Retrieved September 12, 2026, from <https://sabr.org/lahman-database/>

5. Appendix

Data Cleaning Code

```
## Building data frames

pos <- read.csv("FieldingPost.csv")
batting <- read.csv("batting.csv")

stats <- merge(batting, pos[, c("playerID", "POS")],
```

```

    by = "playerID", all.x = TRUE)# Add positions

stats_clean <- stats[stats$lgID == "AL" |
                    stats$lgID == "NL", ] # Focuses on American and National League

stats_clean <- stats_clean[!is.na(stats_clean$POS), ] # Removes batting data without positions

stats_clean$BA <- ifelse(stats_clean$AB > 0, stats_clean$H /
                        stats_clean$AB, NA) ## Adds batting average

stats_wo_pitch <- stats_clean[stats_clean$POS != "P", ] # Removes pitchers for different rules

# league and year
by_league_W_pitch <- aggregate(cbind(G, AB, R, H, HR, RBI, SB, BB, SO) ~ lgID + yearID,
                              data = stats_clean, FUN = sum)

# Adds Batting Average
by_league_W_pitch$BA <- by_league_W_pitch$H / by_league_W_pitch$AB

# league and year no pitcher
by_league_Wo_pitch <- aggregate(cbind(G, AB, R, H, HR, RBI, SB, BB, SO) ~ lgID + yearID,
                               data = stats_wo_pitch, FUN = sum)

# Adds Batting Average
by_league_Wo_pitch$BA <- by_league_Wo_pitch$H / by_league_Wo_pitch$AB # Adds Batting Average

# team and year
by_team <- aggregate(cbind(G, AB, R, H, HR, RBI, SB, BB, SO) ~ teamID + yearID,
                    data = stats_clean, FUN = sum)

# Adds Batting Average
by_team$BA <- by_team$H / by_team$AB # Adds Batting Average

by_year <- aggregate(cbind(G, AB, R, H, HR, RBI, SB, BB, SO) ~ yearID,
                    data = stats_clean, FUN = sum) #year

# Adds Batting Average
by_year$BA <- by_year$H / by_year$AB # Adds Batting Average

# Position
by_pos <- aggregate(cbind(G, AB, R, H, HR, RBI, SB, BB, SO) ~ POS,
                   data = stats_clean, FUN = sum)

# Adds Batting Average
by_pos$BA <- by_pos$H / by_pos$AB

# rename positions
position_names <- c(
  "1B" = "First Base",
  "2B" = "Second Base",
  "3B" = "Third Base",
  "C"  = "Catcher",
  "CF" = "Center Field",
  "LF" = "Left Field",
  "P"  = "Pitcher",
  "RF" = "Right Field",
  "SS" = "Shortstop"
)

by_pos$POS <- position_names[by_pos$POS]

#Position and Team
by_team_pos <- aggregate(cbind(AB, H, HR, BB, SO, BA) ~ teamID + POS, data = stats_clean, FUN = sum)
# Adds Batting Average

```

```
by_team_pos$BA <- by_team_pos$H / by_team_pos$AB
```

```
by_team_pos$Position <- position_names[by_team_pos$POS]
```

Code for Tables

```
# columns of interest
stat_cols <- c("AB", "R", "H", "HR", "SB", "BB", "SO", "BA")

# generates summary statistics
compact_summary <- function(df, cols) {

  variable_names <- c(
    AB = "At Bats",
    R = "Runs",
    H = "Hits",
    HR = "Home Runs",
    SB = "Stolen Bases",
    BB = "Base on Balls",
    SO = "Strikeouts",
    BA = "Batting Average"
  )

  data.frame(
    Variable = unname(variable_names[cols]),
    Minimum = sapply(df[cols], min, na.rm = TRUE),
    Mean = sapply(df[cols], mean, na.rm = TRUE),
    Maximum = sapply(df[cols], max, na.rm = TRUE),
    `Standard Deviation` = sapply(df[cols], sd, na.rm = TRUE),
    row.names = NULL
  )
}

# formats numbers in table
format_table <- function(table_name) {
  table_name[, 2:5] <- lapply(
    table_name[, 2:5],
    function(x) format(round(x, 3), big.mark = ",", trim = TRUE)
  )

  return(table_name)
}

league_summary_W_pitch <- compact_summary(
  by_league_W_pitch,
  stat_cols
)

knitr::kable(
  format_table(league_summary_W_pitch),
  caption = "Statistics Aggregated by League (With Pitchers)"
)

league_summary_Wo_pitch <- compact_summary(
  by_league_Wo_pitch,
  stat_cols
)
```

```
knitr::kable(
  format_table(league_summary_wo_pitch),
  caption = "Statistics Aggregated by League (Without Pitchers)"
)
```

Code for Plots

```
ggplot(by_pos, aes(x = POS, y = BA)) +
  geom_bar(stat = "identity", fill = "maroon") +
  labs(
    title = "Batting Average by Position",
    x = "Position",
    y = "Batting Average"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

```
ggplot(by_team_pos, aes(x = Position, y = SO)) +
  geom_boxplot(fill="maroon") +
  labs(
    title = "Strikeouts Across Teams by Position",
    x = "Position",
    y = "Strikeouts"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

```
ggplot(by_team, aes(x=H, y=HR)) +
  geom_point(col="maroon") +
  labs(
    title="Hits vs Home Runs Across Teams",
    x = "Hits",
    y = "Home Runs"
  )
```

```
ggplot(by_team, aes(x=SO, y=BA)) +
  geom_point(col="maroon") +
  labs(
    title="Strikeouts vs Batting Average Across Teams",
    x = "Strikeouts",
    y = "Batting Averages"
  )
```